

MALUS

above ovate, dark green leaves in late spring. These are followed by long-lasting, rounded, yellow or red crab apples. Z4–8. **‘Red Sentinel’**, H and S 22ft (7m), is broadly upright. Bears white flowers in late spring, and long-lasting, yellow-flushed red, later glossy, dark red fruits. **‘Yellow Siberian’** has white flowers, sometimes pink-tinged, and yellow crab apples. **M. ‘Royalty’** illus. p.84. **M. sargentii**, syn. *M. toringo* subsp. *sargentii*, illus. p.110. **M. x scheidekeri ‘Red Jade’**. See *M. ‘Red Jade’*. **M. sieboldii**. See *M. toringo*. **M. ‘Snowcloud’** illus. p.81. **M. ‘Sun Rival’**. Semiweeping, umbrella-shaped, small tree, with arching branches. H and S 10ft (3m). In mid-spring, clusters of red buds open to white-tinged, pink flowers. Small, round, red fruits last into winter. Very attractive to bees and birds. Z4–8 **M. toringo ‘Scarlett’**. Small, round-headed tree. H and S 20–25ft (6–8m). Has handsome, deeply lobed leaves, which emerge wine-red and age to deep green, reddish beneath; fall color may vary from yellow to orange and red. Bears abundant, mauve-pink flowers, followed by persistent, small, round, dark red fruits. H6. **subsp. sargentii** see *M. sargentii*. **M. toringoides**. See *M. bhutanica*. **M. transitoria**. Deciduous, spreading, elegant tree. H 25ft (8m), S 30ft (10m). Oblong to deeply 3-lobed, mid-green leaves turn yellow in fall. Produces abundant, single, white flowers in late spring, followed by very small, pealike, pale yellow crab apples. Z4–8 **M. trilobata**. Deciduous, conical tree. H 50ft (15m), S 22ft (7m). Has maplelike, lobed, glossy, bright green leaves that often become brightly colored in fall. Bears single, white flowers in early summer, followed by small, rounded or pear-shaped, red-flushed green fruits. Z4–8 **M. tschonoskii** illus. p.77. **M. ‘Van Eseltine’**. Deciduous, upright tree. H 20ft (6m), S 12ft (4m). Has ovate, dark green leaves. Bears double, pink flowers in late spring and rounded, red-flushed, yellow crab apples in fall. Z4–8 **M. ‘Veitch’s Scarlet’** illus. p.89. **M. x zumi ‘Calocarpa’**. Upright, pyramidal to rounded tree. H 28ft (9m), S 25ft (8m). Has ovate, often deeply lobed, dark green leaves. White flowers open from pink buds in late spring, followed by long-lasting, cherrylike, red fruits. Z4–8

MALVA
Mallow

MALVACEAE

Genus of annuals, biennials and free-flowering, short-lived perennials. Fully hardy. Requires sun and fertile, moist but well-drained soil. Propagate species by seed in early spring or early summer, selected forms by cuttings from firm, basal shoots in late spring or summer. These shoots may be encouraged by cutting the plant back after first flowers have faded. **M. moschata** (Musk mallow) illus. p.232. **f. alba** is an upright, bushy, woody-based perennial. H 3ft (90cm), S 2ft (60cm). Has

slightly musk-scented leaves; lower ones heart-shaped, upper ones finely divided. In summer, bears axillary clusters of saucer-shaped, white flowers. Z3–8

Malvastrum capensis. See *Anisodonteia capensis*.

MALVAVISCUS

MALVACEAE

Genus of evergreen shrubs and trees, grown for their flowers. Frost tender. Needs a site in full light and fertile, well-drained soil. Water containerized plants freely during growing season, moderately at other times. To maintain shape, flowered stems may be cut back hard in late winter. Propagate by seed in spring or by semiripe cuttings in summer. Whitefly and red spider mite may be troublesome. **M. arboreus** (Sleepy mallow). Vigorous, evergreen, rounded shrub. H to 12ft (4m) or more, S 5–10ft (1.5–3m). Serrated, bright green leaves are soft-haired. Has bright red flowers with protruding stamens in summer and fall. Z10–11

MAMMILLARIA

CACTACEAE

Genus of spherical to cylindrical or columnar cacti, grown for their funnel-shaped flowers that develop mainly in rings around the crowns. Flowers, offsets, and long, slender to spherical seed pods grow between tubercles on a spiny, green stem with extended areoles. Frost tender. Requires full sun and very well-drained soil. Keep completely dry in winter, otherwise plants rot easily. Propagate by seed in late winter or early spring. **M. bocasana** (Snowball cactus) illus. p.480. **M. candida**, syn. *Mammilloidia candida* (Snowball cushion cactus). Slow-growing, solitary or clump-forming, perennial cactus. H and S 6in (15cm). Columnar, green stem is densely covered with short, stiff, white spines. Bears white-edged, rose flowers, ½–¾in (1–2cm) across, in spring. Water sparingly in summer. Z10–11 (min. 41°F/5°C) **M. centricirra**. See *M. magnimamma*. **M. conoidea**. See *Neolloydia conoidea*. **M. dealbata**. See *M. haageana*. **M. densispina**. Slow-growing, solitary perennial cactus. H and S 4in (10cm). Spherical or cylindrical, dark green stems have white-woolly areoles with yellow or pale brown radial spines and 5 or 6 reddish brown, black-tipped centrals. From spring to summer, bears sulphur-yellow flowers, often with red-flushed outer petals, ¾in (2cm) long. Z10–11 (min. 34°F/1°C) **M. elongata** (Gold lace cactus) illus. p.494. **M. geminispina** illus. p.480. **M. gracilis**. See *M. vetula* subsp. *gracilis*. **M. haageana**, syn. *M. dealbata*. Spherical to columnar, perennial cactus, offsetting from the base and sides. H 1ft (30cm), S 8in (20cm). Spherical to cylindrical, mid-green stems have short, thin, white radial spines and longer, black-tipped, red-brown centrals. Bears bright red flowers. ½in (1cm) across, in spring. Z10–11 (min. 34°F/1°C) **M. hahniana** (Old lady cactus) illus. p.479. **M. magnimamma**, syn. *M. centricirra*.

Very variable, clump-forming, perennial cactus. H 1ft (30cm), S 2ft (60cm). Gray-green stems have pronounced, angular, dark green tubercles with white spines of variable length. Bears cream, pink, or red flowers, in spring and possibly again in late summer. Z10–11 (min. 34°F/1°C) **M. microhelia** illus. p.491. **M. plumosa** illus. p.483. **M. prolifera** (Strawberry cactus). Clump-forming, perennial cactus. H 4in (10cm), S 1ft (30cm). Green stem have dense, golden to white spines. Bears abundant, cream or yellow flowers, 1–2cm (½–¾in) across, in summer, followed by edible red berries that taste like strawberries. Z10–11 (min. 1°C/34°F) **M. rhodantha**. Spherical to columnar, perennial cactus. H and S 2ft (60cm). Green stem, branching from crown with age, is densely covered with white to brown and yellow spines, often curved. In summer, produces purplish-pink flowers, ¾in (2cm) long. Z10–11 (min. 34°F/1°C) **M. schiedeana**. Solitary or clump-forming, perennial cactus. H 4in (10cm), S 1ft (30cm). The green stem is covered with short, feathery, yellow spines that turn white. Produces cream flowers and narrow, red seed pods from summer to fall. Z10–11 (min. 34°F/1°C) **M. sempervivi** illus. p.484. **M. vetula subsp. gracilis**, syn. *M. gracilis*. Clump-forming, perennial cactus. H 2in (5cm), S 8in (20cm). Produces a columnar, green stem, densely covered with brown and yellowish-white spines. From spring to summer, bears pale cream flowers, ½–¾in (1–2cm) across. Stem is shallow-rooted and reroots readily. Z10–11 (min. 34°F/1°C). **var. fragilis**, H 1½in (4cm), is more fragile and has off-white spines. **M. zeilmanniana** (Rose pincushion) illus. p.486.

Mammilloidia candida. See *Mammillaria candida*.

MANDEVILLA
SYN. DIPLADENIA

APOCYNACEAE

Genus of evergreen, semievergreen, or deciduous, woody-stemmed, twining climbers, with some perennials, grown for their large, trumpet-shaped flowers. Frost tender. Grow in any well-drained soil, with light shade in summer. Water freely in full growth, sparingly at other times. Provide support; thin out and spur back congested growth in early spring. Propagate by seed in spring or by semiripe cuttings in summer. Whitefly and red spider mite may cause problems. ☉Contact with the sap may cause skin irritation, and all parts may cause mild stomach upset if ingested. **M. x amabilis ‘Alice du Pont’**, syn. *M. x amoena* ‘Alice du Pont’, illus. p.461. **M. x amoena ‘Alice du Pont’**. See *M. x amabilis* ‘Alice du Pont’. **M. boliviensis**, syn. *Dipladenia boliviensis*. Vigorous, evergreen, woody-stemmed, twining climber. H to 12ft (4m). Oblong, pointed leaves are lustrous green. Has large, trumpet-shaped, white flowers with gold eyes in small clusters in summer. Z10–12 (min. 50°F/10°C) **M. laxa**, syn. *M. suaveolens*, *M. tweediana*

(Chilean jasmine). Fast-growing, deciduous or semievergreen, woody-stemmed, twining climber. H 15ft (5m) or more. Ovate leaves have heart-shaped bases. Clusters of fragrant, white flowers open in summer to early fall. Z10–11 (min. 41°F/5°C) **M. splendens**, syn. *Dipladenia splendens*, illus. p.460. **M. suaveolens**. See *M. laxa*. **M. tweediana**. See *M. laxa*.

MANDRAGORA
Mandrake

SOLANACEAE

Genus of rosetted perennials with large, deep, fleshy roots. Frost hardy. Requires a position in sun or partial shade, and deep, humus-rich, well-drained soil. Resents being transplanted. Propagate by seed in fall. ☉Alkaloids in the plant may be harmful if ingested. **M. officinarum** (Common mandrake, Devil’s apple) illus. p.356.

MANETTIA

RUBIACEAE

Genus of evergreen perennials and woody-stemmed, twining climbers, grown for their small, showy flowers. Frost tender. Grow in humus-rich, well-drained soil, with partial shade in summer. Water regularly, sparingly in low temperatures. Stems need support. Cut back if required in spring. Propagate by softwood or semiripe cuttings in summer. Whitefly is sometimes a problem. **M. bicolor**. See *M. luteorubra*. **M. cordifolia** (Firecracker vine). Fast-growing, evergreen, soft-stemmed, twining climber. H 6ft (2m) or more. Has oblong, lance-shaped or narrowly heart-shaped, glossy leaves. Funnel-shaped, red flowers, sometimes yellow-flushed on the lobes, are borne in small clusters in summer. Z10–12 (min. 45°F/7°C) **M. inflata**. See *M. luteorubra*. **M. luteorubra**, syn. *M. bicolor*, *M. inflata* (Brazilian firecracker). Evergreen, fast-growing, semiwoody-stemmed, twining climber. H 6ft (2m). Has slightly sticky, hairy stems and glossy leaves. Produces small, funnel-shaped, red flowers, with yellow tips, in summer. Z10–12 (min. 45°F/7°C)

Manfreda maculosa. See *Agave maculosa*. **Manglietia**. See *Magnolia*. **Manglietia fordiana**. See *Magnolia fordiana*. **Manglietia insignis**. See *Magnolia insignis*.

MARANTA

MARANTACEAE

Genus of evergreen, rhizomatous perennials, grown for their distinctively patterned, colored foliage. Frost tender. Needs constant, high humidity and a shaded position away from drafts or wind. Grow in humus-rich, well-drained soil. Propagate by division or by stem cuttings in spring. **M. leuconeura** (Prayer plant). **‘Erythroneura’** (syn. *M.I. ‘Erythrophylla’*; Herringbone plant) illus. p.475.